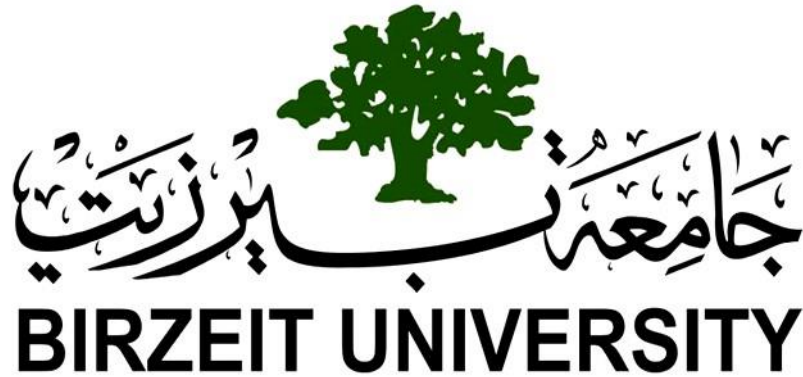




Faculty of Engineering & Technology

Electrical & Computer Engineering Department

CIRCUITS AND ELECTRONICS LABORATORY ENEE2103

Report#2

Experiment 3: First and Second Order Circuit

Prepared by:

Ahmed Zubaidia 1200105

Partners:

Jana Herzallah 1201139

Lana Badwan 1200071

Jana AbuNasser 1201110

Instructor: Dr.Mohammad Jihad Al Ju'Beh

Teaching Assistant: Eng.Shadi

Section: 1

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Abstract

This experiment aims to evaluate the results and findings regarding analyzing and testing time responses of three circuits (RL, RC and second order RLC). It also investigates how the initial state of the dynamic elements influences time response. Also, the real values should be analyzed and rated compared with the theoretical values. In addition, it attempts to find the circuit first and second order parameters according to the circuit response. Electric value measurement with Oscilloscope. We consider the classification of the present response in a series RLC circuit, namely overdamped, underdamped and critically damped.

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Theory

RC Circuit

An RC circuit, a fundamental electrical device, consists of a resistor (R) along with a capacitor (C) in series. This circuit is used in different electronic products, like camera flashes timing mechanism or pacemaker control circuits. In an RC circuit, the resistor controls the speed at which the capacitor charges and also discharges. One of the most important property of an RC circuit will be the time constant (τ), and that is the result of the resistance (R) in ohms and the capacitance (C) in farads ($\tau = R C$) [1].

The behavior of the RC circuit is mostly controlled by the charging and discharging curves of the capacitor. Once voltage is used, the capacitor charges gradually but exponentially. It charges fast initially but slows dramatically near its maximum charge. This particular charging process could be mathematically referred to as $V_c = V_s (1 - e^{-t / RC})$, where V_c is the capacitor voltage at moment t , V_s is the source voltage, and e will be the foundation of the natural logarithm. In practice, an RC circuit is considered steady after about 5, once the capacitor is almost fully charged. This particular property makes RC circuits useful for electronic timing and waveform shaping applications like filters and oscillators where time constants need to be precisely controlled [1].

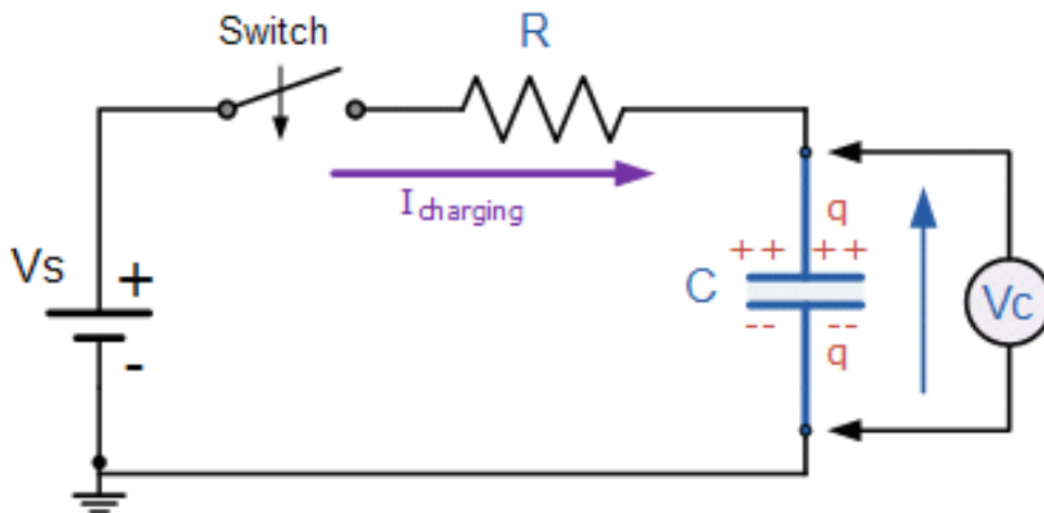


Figure 1: RC Circuit

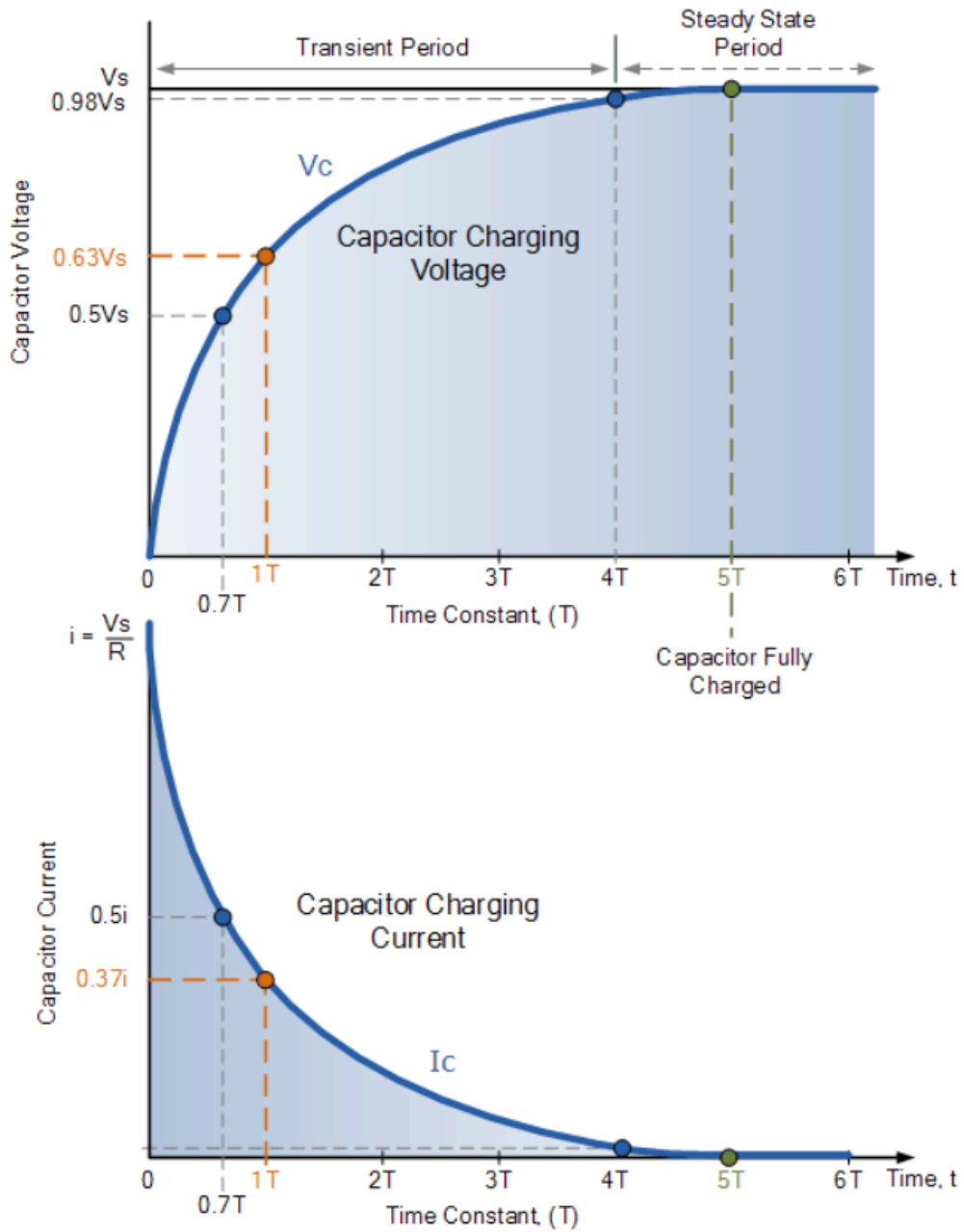


Figure 2: RC Charging Circuit Curves

During the charging process of an RC circuit, the time constant is achieved when the voltage level reaches 63.2% of its maximum supply voltage (V_{max}). Instead, in discharging, the time constant is set at the low voltage of 36.8% of V_{max} . This distinction reflects the different behaviors of capacitor in accumulating and releasing charge in an RC circuit [1].

RL Circuit

In a circuit where an inductor and a resistor are attached in series, both elements flow current when voltage is applied. But because they are in series, the current that flows through each is the same. The current through the inductor is called I_L , and the current through the resistor is called I_R . The impedance values of the resistor and inductor affect the I_R and I_L magnitudes respectively [2].

The operational characteristics of an RL circuit are determined by resistor resistance (R), inductor inductance (L), and applied voltage frequency. The time constant for an RL circuit, represented by, will be the ratio of inductance to resistance ($= L/R$), which determines the speed of current evolution of the circuit [2].

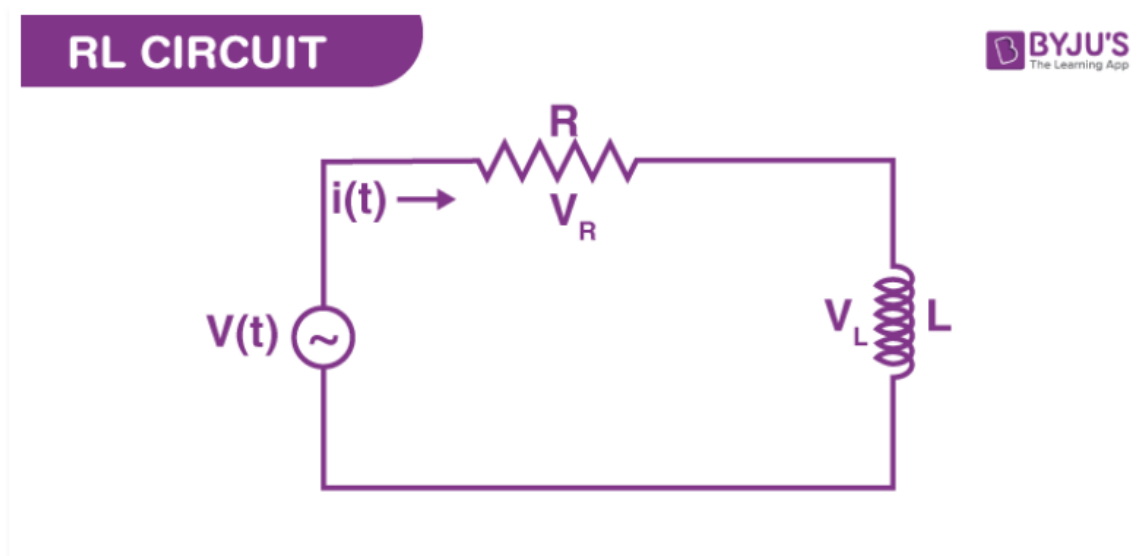


Figure 3: RL Circuit

In an RL circuit where the behavior depends on the resistance (R), inductance (L) and the frequency of the applied voltage, the time constant is very important. This time constant is equal to the ratio of inductance to resistance $= L/R$. It is important because it sets the rate at which the circuit current changes. The key formulas for calculating the specific dynamics of the inductor in an RL circuit are given below $\therefore$ To find the voltage across the inductor and the current in any particular time we can use these equations $v(t) = Vin(1 - e^{(-t * R)/L})$ And for current $I(t) = I_0 * \left(1 - e^{\frac{-t * R}{L}}\right)$ [2].

RLC circuit

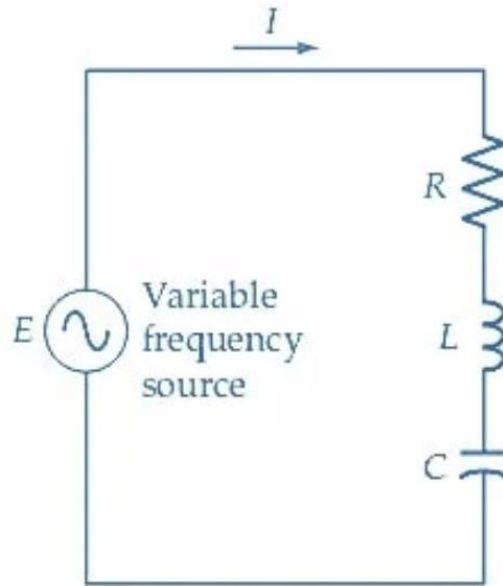


Figure 4:RLC Circuit

An RLC circuit, provided by its name, includes a resistor (R), a capacitor (C), and also an inductor (L), organized in sequence or parallel configurations. This kind of circuit is important to design an oscillator circuit which is utilized in TVs and radio receivers. In analog applications also, RLC circuits are used as damper circuits. In an RLC circuit, if the switch is closed, the charge in the capacitor moves toward the inductor and the inductor charges up. After the capacitor is totally discharged, the inductor charges the charge back into the capacitor. This back-and-forth charges between the inductor and capacitor continues but the overall charge decreases gradually due to some loss in the process, eventually reaching zero [3].

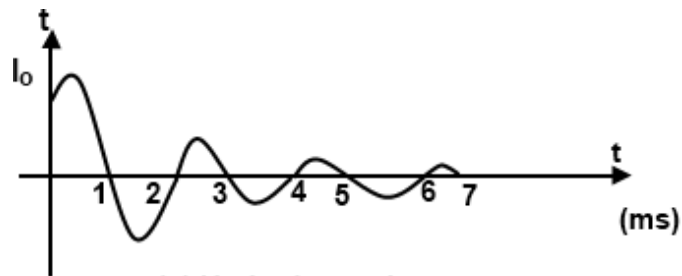
The basic characteristics of an RLC series circuit are:

- Current through all components is the same, but voltage drops across each component are out of phase.
- The voltages throughout each component depend upon the circuit present and the values of R, L, and C [3].

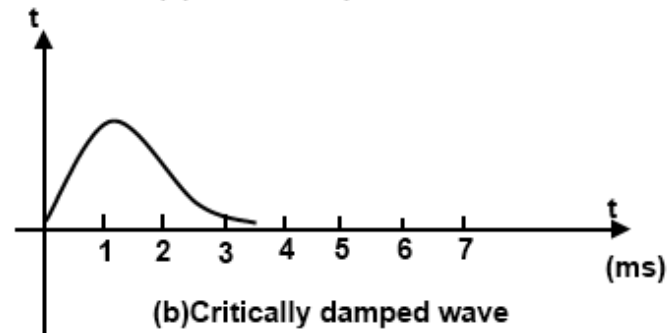
The dynamics of the RLC circuit is defined further by its damping coefficient and resonant frequency. The damping coefficient (α) is given by $\alpha = R/2L$, which influences the behavior of the circuit oscillations in time. The resonant frequency (ω) $= 1/LC$ is the frequency of oscillation of the circuit [3].

According to the relationship of resistance, inductance, and capacitance the current in the circuit can act like the following:

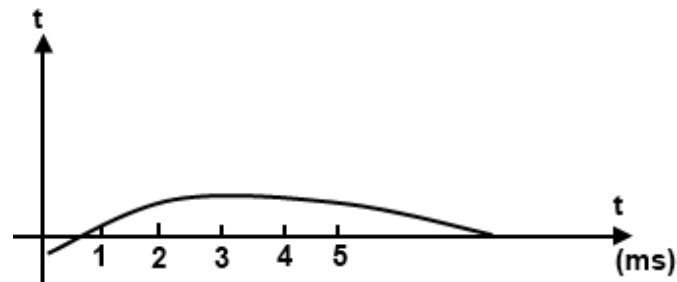
- Over-damping ($\alpha > \omega$): where the system resumes equilibrium without oscillation.
- Critical damping ($\alpha = \omega$): the system returns to equilibrium as quickly as possible without oscillation.
- Under-damping ($\alpha < \omega$): the system oscillates, and the amplitude gradually gets zeroed out [3].



(a) Underdamped wave



(b) Critically damped wave



(c) Overdamped wave

Figure 5: cases of Damping of RLC

Procedure, Calculations and Discussions

RC Circuit

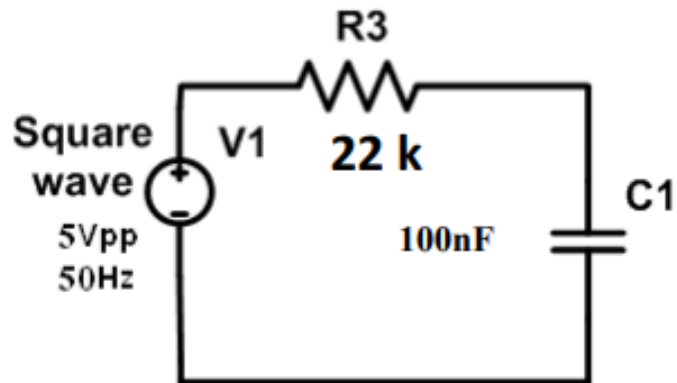


Figure 6: RC circuit

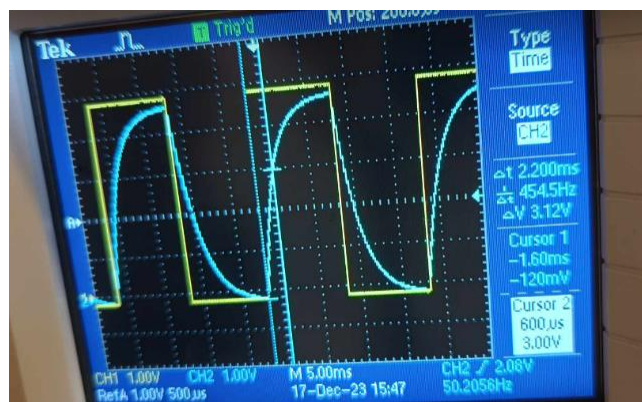


Figure 7: time constant τ

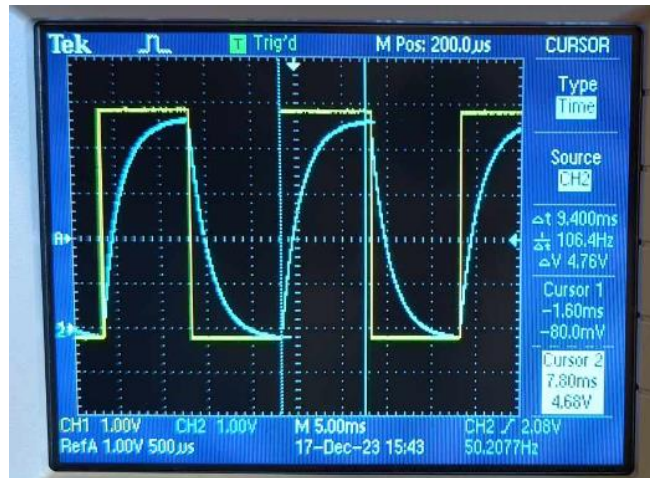


Figure 8: Voltage on capacitor

The time constant was $\tau = 2.2$ ms in the experiment. With this time constant we estimated the capacitance as about 100 nF by the equation $C = \frac{\tau}{R} = \frac{2.2}{22k} = 100nf$. This is consistent with what we expected and shows that our experiment and calculation were correct. The 2.2 ms time constant was agreed upon by both experiment and theory.

We also measured the steady-state voltage $V_c = (4.68 \text{ volts})$ on the capacitor. it is important because it shows the last behavior of capacitor in the circuit.

We can say, the experiment showed the operation of an RC circuit. Our measurement agreement with the theoretical values confirms that the experiment was correctly done. and we understand the circuit well.

RL Circuit

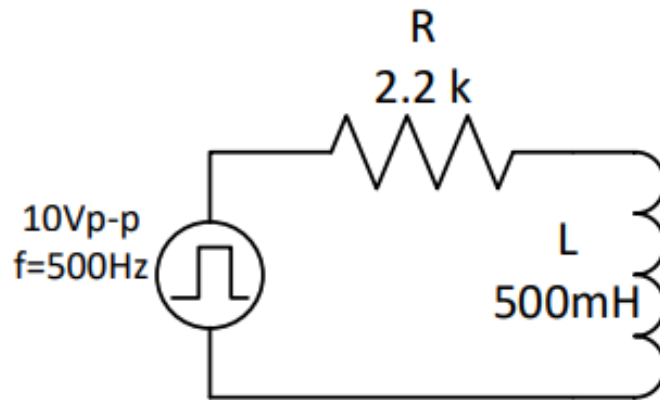


Figure 9: RL circuit

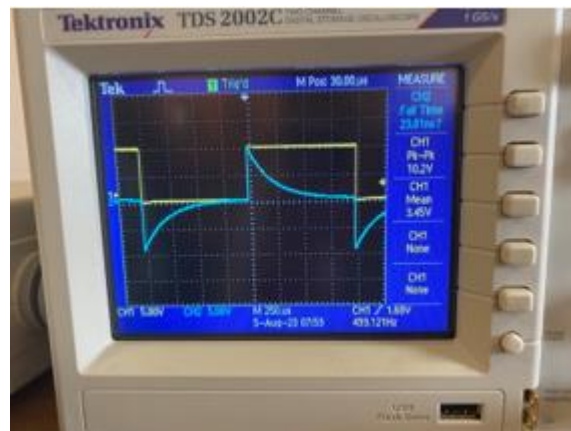


Figure 10: RL Circuit simulation on the oscilloscope

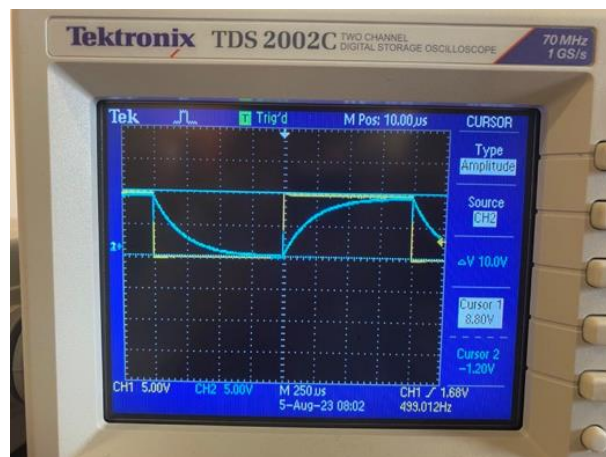


Figure 11: voltage on inductor

We can see that the voltage on $V_L = \Delta v = 10$ volt

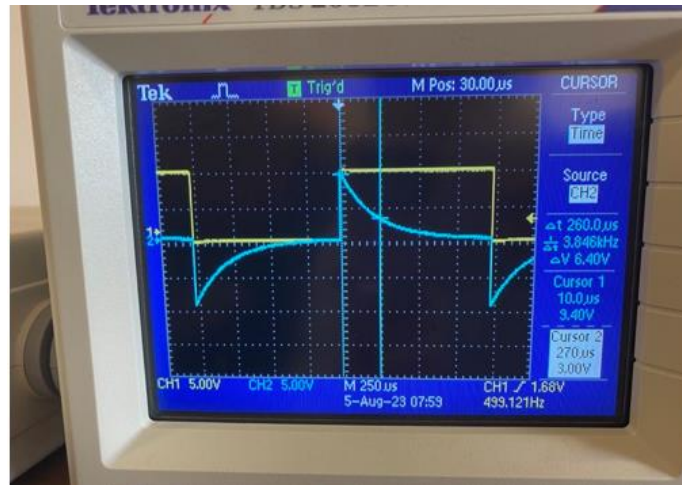


Figure 12: : Time constant (τ)

For the time constant in this experiment, it was $\tau = 0.26$ ms by experimental means. This was close to the theoretical time constant $\tau = \frac{L}{R} = \frac{500 \text{ mH}}{2.2 \text{ K}} = 0.227 \text{ ms}$. Also the current in the circuit was experimentally calculated as 4.55 mA by $\frac{V_L}{R} = \frac{10}{2.2 \text{ k}} = 4.55 \text{ mA}$.

In addition, the period of the square wave was set to twice the time constant and a new period T was obtained ($2 * 0.26 \text{ ms}$).

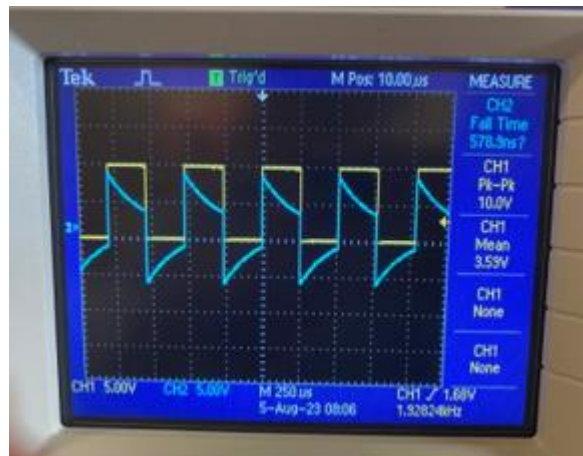


Figure 13: The voltage response after changing the period time

The waveform observed with T (the period of the square wave) set to twice the time constant (2τ) reveals some characteristic of the inductor behavior in the circuit. At this value, the inductor it can't reach to its maximum current (I) in the charging , nor can the inductor fully discharge to its minimum current (0 A) in the discharging.

This occurs because the inductor resists current changes. When the time constant is large part of the period, the inductor's current doesn't reach the maximum or minimum values it would have reached with more time or a shorter time constant. It shows how the inductive circuit reacted to voltage changing and shows the practical usage of time constant in circuit design and analysis. it shows the way the inductor's charging and discharging speeds are limited by time available in each cycle of the waveform.

RLC Circuit

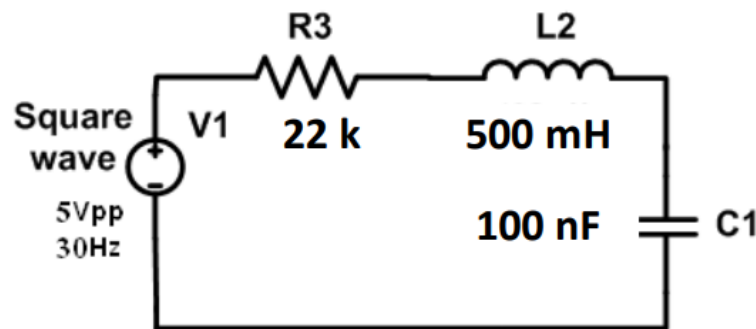


Figure 14: RLC Circuit

Response type



Figure 15: RLC circuit input and output voltage

In order to determine the type of response, the signal generator is configured to produce a square waveform that repeats regularly, along with a direct current offset of 2.5 volts. This waveform has a frequency of 30 Hz.

The equation was used to calculate the value of R, which will result in a response of critical damping.

- $R_{\text{critical}} = \frac{1}{\sqrt{LC}} = 4.472 \text{ K } \Omega$
- $\alpha = \frac{R}{2L} = 22 \text{ K}$

Response type is Over Damping because $\alpha > \omega$ since $(\frac{R}{2L} > \sqrt{\frac{1}{LC}})$.

The resistance required to meet the standards for critical damping and under damping was determined by substituting R3 with a resistance decade box. To detect a shift in the type of response, we gradually adjusted the variable resistor.

The experimental value of R3 for achieving critical damping is 4471 Ω .

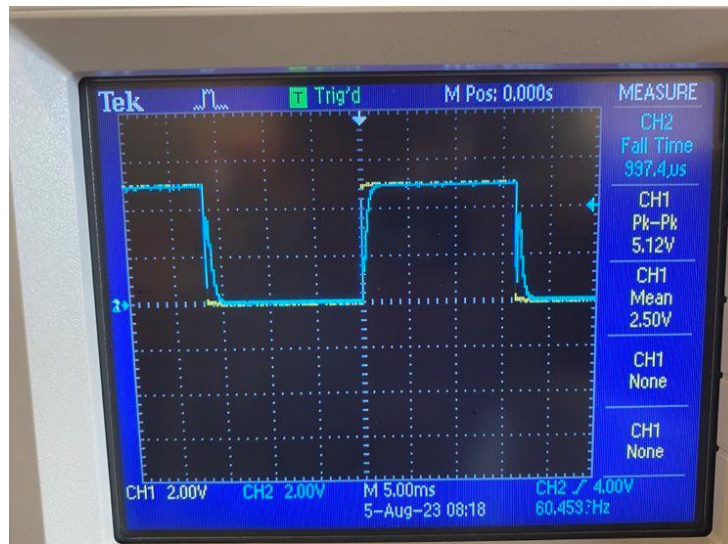


Figure 16: Critical damping $R_3=4471\ \text{ohm}$

Now we this is R_3 under-damping = $500\ \Omega$



Figure 17: Underdamping $R_3=500\ \text{ohm}$

R_3 for over damping = $22\text{ k}\Omega$

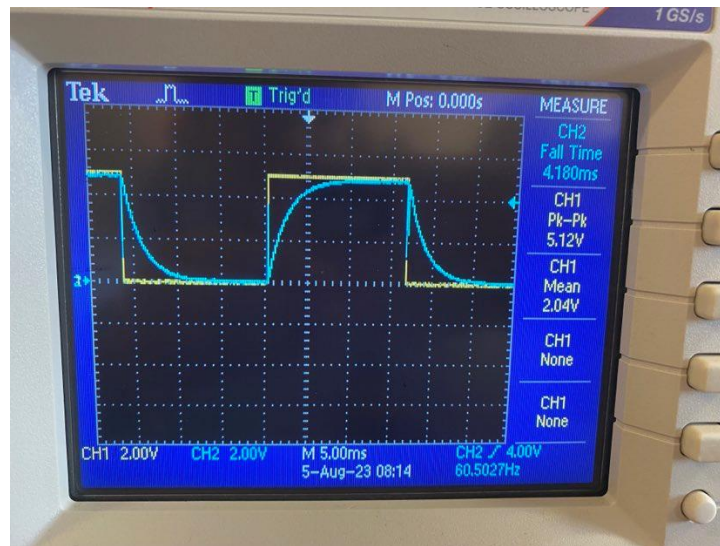


Figure 18: Overdamping $R_3=22\text{ k ohm}$

Response parameters

The circuit keep as its connected but some values are changed:

$R = 500\text{ohm}$, $C = 200\text{nF}$, $L = 250\text{mH}$



Figure 19: The under-damped response used to measure damping coefficients

I Used these equations to find the coefficients:

1. Decay time constant $\tau = \frac{T_b - T_a}{\ln\left(\frac{v_a - v_0}{v_b - v_0}\right)}$
2. Damping Coefficient $\alpha = 1/\tau$
3. Damped radian frequency $\omega_d = 2\pi / (T_b - T_a)$

With R = 500Ω:

- Time Constants: $t_a=0.707\text{ms}$, $t_b=2.1203\text{ms}$
- Voltage Values: $V_a=8.5\text{V}$, $V_b=6.7\text{V}$, $V_0(\infty)=5\text{V}$
- Time Constant (τ): $\tau=2\text{ms}$
- Damping Coefficient (α): $\alpha=511.06$
- Damped Natural Frequency (ω_d): $\omega_d=4446.69 \text{ rad/s}$

When C increase τ increases too, but α and ω_d reduced.

- Time Constants: $t_a=1.0063\text{ms}$, $t_b=3.0183\text{ms}$
- Voltage Values: $V_a=8.0234\text{V}$, $V_b=6.1055\text{V}$, $V_0(\infty)=5\text{V}$
- Time Constant (τ): $\tau=0.002\text{s}$
- Damping Coefficient (α): $\alpha=499.9$
- Damped Natural Frequency (ω_d): $\omega_d=3122.85\text{rad/s}$

- **Inductance (L): 250 mH**

Time Constants:

$t_a=0.503\text{ms}$

$t_b=1.5093\text{ms}$

Voltage Values:

$V_a = 8.0234\text{V}$

$V_b = 6.1055\text{V}$

$V_0(\infty)=5\text{V}$

Time Constant (τ): $\tau = 0.001 \text{ s}$

Damping Coefficient (α) $\alpha = 999.785$

Damped Natural Frequency $\omega_d = 6243\text{rad/s}$

When the inductance (L) of an RLC circuit is reduced, the time constant (τ) almost remains unchanged, indicating that the circuit response time to changes is not greatly affected by smaller inductance values. However, the damping coefficient (α) and damped natural frequency (ω_d) both decrease. This reduction in α and ω_d indicates the circuit is less damped and oscillates at a lower frequency which may be due to the reduced ability of the inductor to store energy.

In the opposite direction, if C is increased, τ increases likewise. A larger capacitance means it takes longer to charge and discharge the capacitor, thus the circuit response time is longer. At the same time, α and ω_d decrease, similar to when L is reduced. The lower α indicates less damping, possibly because the increased capacitance allows more charge storage, leading to sustained oscillations. The decrease in ω_d with higher capacitance implies lower oscillation frequency since the circuit takes longer to complete each cycle due to the larger capacitor.

Conclusion

We learned how to apply the concepts of first and second order linear circuits in real-life scenarios. Additionally, we examined the behavior of current and voltage in these circuits and calculated their time constant using oscilloscope cursors. We then compared our experimental findings with the expected outcomes based on theoretical predictions. We could see the dampening cases. Modifying the capacitor value within the RLC circuit is as simple as changing the capacitor value. We also learned how the circuit behavior would change in case the component values have been altered. We made adjustments to the settings of the inductor and capacitor. In addition, we acquired knowledge about the three types of response states, namely over damping, crucial damping, and damping. and How can I determine some parameters using the graphs.

References

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- [2] "2," [Online]. Available: <https://byjus.com/jee/rl-circuit/>.
- [3] "3," [Online]. Available: <https://eepower.com/technical-articles/determining-series-rlc-circuit-resonance-frequency/#>.

ENEE2103

Circuits & Electronics Lab

ENEE2103

Experiment#3 First and Second Order Circuit

Objectives:

1. To use the Oscilloscope to measure electric values.
2. To test and analyze the time responses of RL and RC circuits.
3. To test and analyze the time response of the second order RLC circuit.
4. To test the effect of the initial state of the dynamic elements on the time response.
5. To determine the first and second order circuits parameters from the circuit response.

Equipment:

1. Digital Multimeter.
2. Oscilloscope (TDS-2002B).
3. Power supply.
4. Signal generator.
5. Discrete Capacitors and Resistors, Inductance decade box, Resistance decade box

Pre-lab:

3. Simulate the circuits in the procedure section and determine the required values (set the parameters that must be assigned by the instructor in the procedure to proper values).
4. Verify if Simulation Results match the expected results

Procedure:

A. RC Circuit:

1. Set the resistance Decade Box to the value of R3 then measure Value of C1 using the RLC meter.
2. Connect the circuit of Fig (3.1)
3. Set the signal generator to square wave 5Vp-p and 50Hz with dc offset=2.5V
4. Connect the Oscilloscope to the Capacitance terminals.
5. From the response displayed on the oscilloscope, determine the value of the system time constant.
6. Determine the steady state voltage value on the capacitor.
7. Use your measurements of the time constants to determine the value of the capacitance.

Handwritten notes:

$V_{max} = 4.6V$

$\tau = 4.6 * 0.636 = 2.898$

$V_{max} = 4.68 \Rightarrow 0.63 V_{max} = 2.99 \Rightarrow \tau = 2.2ns$

Handwritten calculations:

$t =$

Handwritten notes: Measure, Oscilloscope

Fig (3.1)

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0.17
0.24

B. RL Circuit: $V_L = 10 \text{ volts}$ $I = 4.55 \text{ mA}$

1. Connect the circuit of Fig (3.2)
2. Set the signal generator to generate a periodic square waveform with 10Vp-p and frequency=500Hz, dc offset=5V.
3. Connect the oscilloscope to display the voltage response of the inductor.
4. Measure the time constant of the circuit and the steady state values of the voltage and current responses.
5. Determine the behavior of the voltage and current responses in relation to the element characteristic equation.
6. Change the period of the periodic square wave to $T=2\tau_L$ and display the result.
7. Write your conclusion about the displayed waveform.

 $\tau = 960 \mu s$

$\tau = 500 \mu s$
 $= 0.5 \text{ ms}$

 $V_R = 10.3 \text{ volts}$

divide by R
to get the inductor
current.

10Vp-p
f=500Hz

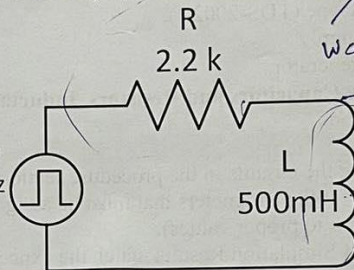


Fig (3.2)

WAVE

$$I = \frac{V_R}{R} = \frac{10.3 \text{ volt}}{R}$$

to get
the i wave

C. RLC Circuit:**I. Response type:**

1. Connect the circuit of Fig (3.3)

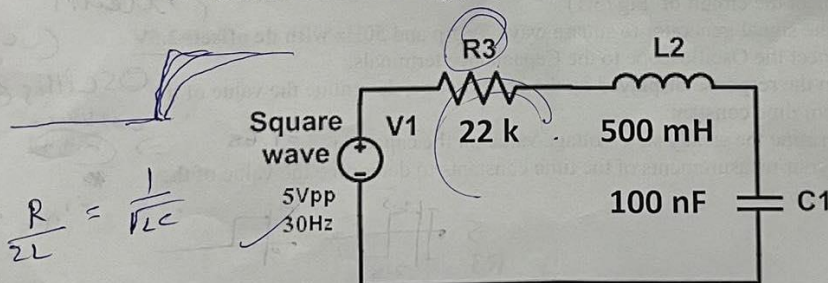


Fig (3.3)

2. Connect the oscilloscope to measure the voltage in the capacitor.
3. Set the signal generator to generate a periodic square waveform with ± 2.5 volts and 30Hz, dc offset=2.5V.
4. Determine the type of the response. *overdamp*
5. Replace R3 by resistance decade box *critical damp* $R3 = ? = 4471 \Omega$
6. Calculate the value of the resistance that satisfied the critical damping and the under damping conditions. $\rightarrow R = 500 \Omega$
7. Change the variable resistor with steps so that you can detect a change in the type of the response. $R_{over} = 22 \text{ k}$

$\frac{R}{2L} = \frac{1}{\tau LC}$
critical

8. Refine your steps around the value for which the transition occurs so that to detect the transition point.
9. Determine the type of the response in each case.

II. Response parameters:

1. Set the value of R_x to (define value through test and confirm it with lab instructor) so that to get an under damped response
2. Use the cursor to measure the decay-envelope time constant (τ), the damping coefficient (α) and the damped frequency (ω_d) as shown in Figure 3.3.
3. Double the value of $C1$ and Measure the parameters defined in step2 noting the effect.
4. Reset the capacitance to its initial value
5. Reduce value of $L2$ to half its value and note the effect on previous parameters.

قسط
بقل

critical = 3220

above critical $\Rightarrow$ overdamped
under " $\Rightarrow$ underdamped

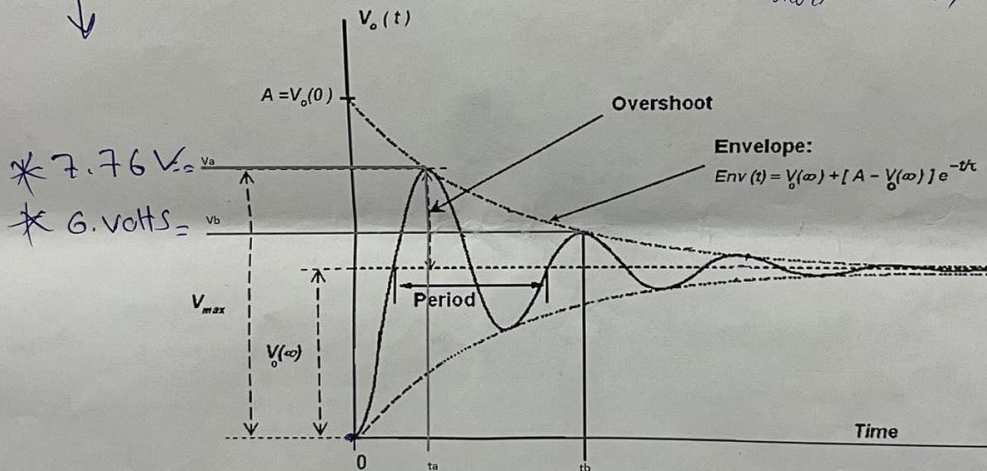


Figure 3.3

$$\tau = \frac{t_b - t_a}{\ln \left(\frac{V_a - V_o(\infty)}{V_b - V_o(\infty)} \right)}$$

$$* V_o(\infty) = 4.88$$

Decay time constant

Damping Coefficient

$$\alpha = \frac{1}{\tau}$$

Damped radian frequency

$$\omega_d = \frac{2\pi}{t_b - t_a}$$

$$* t_a = 440 \mu s$$

Components List:

100nF (2)
Resistance Decade Box
Inductance Decade Box

22kohm, 2.2kohm

$$* \tau = 1.84 ms$$

$t_p =$